

AI-Buddies in MMORPGs: Player Perceptions of Conceptual LLM-Driven NPCs in World of Warcraft

Rasmus Ploug & Marco Scirea

Motivation



LLMs are entering games but **unclear how to** integrate them in MMORPGs



NPC dialogue → currently **static, repetitive**



Research Questions:

RQ1: How do Massive Multiplayer Online Role-Playing Game (MMORPG) players react to the concept of LLM-based agents?

RQ2: What features are players interested in?



Background: WoW and NPCs

- **WoW (2004–today):** one of the largest MMORPGs, built on NPC-driven quests and lore
- **Early design:** teamwork essential; even elites required groups
- **Recent trend:** shift toward **solo-friendly content** with weaker mobs + auto matchmaking
- **New features:** *Follower Dungeons* and *Delves* introduce AI-controlled companions
- This reliance on AI sets the stage for **LLM-driven NPCs**



The AI-Buddy Concept

- **Personal companion NPC** chosen by the player
- Powered by an **LLM for open-ended dialogue** and dynamic interactions
- Can **fill roles** (tank, healer, DPS) and adapt to player behavior
- **Customizable** appearance, personality, and gear
- Adds memory of past interactions → more immersive and personal experience

Method

- **Mixed-method survey** with 273 WoW players
- Used **mock-up screenshots** to illustrate potential AI-Buddy features
- Recruitment: Discord, Reddit, Facebook WoW communities
- **Quantitative:** 36 Likert-scale items + stats (Mann-Whitney U)
- **Qualitative:** two open-ended responses, thematic analysis



Participants

273 players: 86% male, 71% EU / 29% NA

Playstyle: 21% Casual, 18% Hardcore, 62% Mid

Roles: 82% DPS, 33% Healer, 25% Tank

Social: 81% in a guild, 63% with 3+ active in-game friends

Table 3: Comparison of Significant Preferences between Hardcore and Casual Players ($p < 0.05$)

Hardcore Players:

- Value group play more.**
 - Prefer to play with friends other random people.**
 - Concerned about AI-Buddies in competitive content.*
 - Prefer to play with friends over AI-Buddy.
 - Enjoy WoW significantly more.
-

Casual Players:

- More likely to use AI-Buddy when friends are offline.**
 - Prefer solo content over multiplayer content.**
 - Enjoy the shift towards more solo content.**
 - Find AI companions in *Follower Dungeons* more realistic*
 - Find AI companions in *Delves* more realistic*
 - Find *Follower Dungeons* to be more enjoyable.*
 - Find *Delves* to be more enjoyable.*
 - Believe AI-Buddy enhances immersion more.
-

*Significance Levels: * = $p < 0.01$, ** = $p < 0.001$.*

Key Quantitative Results

→ Casual vs Hardcore:

- Casual → more positive, use AI-Buddy when friends offline
- Hardcore → value group play, skeptical of AI in multiplayer

→ **Most desired features:** memory, adaptive learning, multiple roles, customization

→ **Disliked:** AI in PvP (low support)

→ Overall perception → **positive**, especially for solo content

Key Qualitative Results

Concerns:

- Balance & fairness (gear, farming, PvP)
- Reduced **social interaction** in an MMORPG
- Technical issues (lag, server strain)

Suggestions:

- Use player's **alts as AI-Buddies**
- Strong **customization** (name, race, personality)
- Restrict to **solo / non-competitive** content

Table 4: AI-Buddy Features Sorted by Popularity

Highly Liked ($4.5 \leq M \leq 5.0$)

1. Clearly Distinguishable from Human Players
 2. Customizable Appearance (Gear, Looks, Race)
 3. Ability to Perform Multiple Roles (Tank, Healer, DPS)
-

Liked ($3.5 \leq M \leq 4.4$)

4. Memory of Past Interactions
 5. Learns from Player Behavior and Improves Over Time
 6. Ability to Share Stories from World of Warcraft Lore
 7. Reacts to In-Game Events (e.g., Zone Changes)
 8. Limited to Solo or Non-Competitive Group Content
 9. Assists with Dynamic World Events (e.g., World Quests)
 10. Customizable Personality
 11. Helps with Repetitive Content
 12. Suggests In-Game Activities Based on Player Interests
 13. Provides Feedback After Completing Difficult Content
-

Neutral ($2.5 \leq M \leq 3.4$)

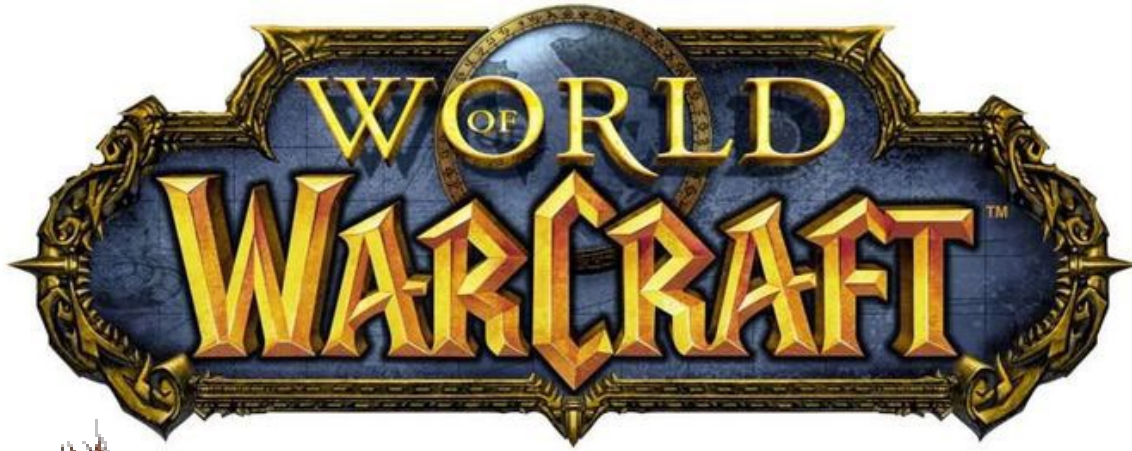
14. Initiates Conversation When Player is Idle
 15. Participates in Role Play
-

Disliked ($1.5 \leq M \leq 2.4$)

16. Assists in World PvP
-

Discussion & Implications:

- **General support**, especially from Casual players
- AI-Buddy aligns with **complementary CA-bots** (enhancing solo play)
- Risks: undermining **social play** and in-game economy
- Strong demand for **memory, adaptation, and customization**
- Careful limits needed → **no PvP, no competitive conten**



Conclusion

- Players see **AI-Buddies** as valuable for **solo play & immersion**
- **Casuals** enthusiastic, **Hardcore** cautious about balance
- Must remain **distinct from human players**
- Avoid PvP / economy disruption → keep scope limited
- Future work: **real MMORPG testing** across different communities

Thank you